QML on TR1 South: summary

Weak Lensing QML pipeline

October 22, 2026

What we did

- Applied the QML power-spectrum estimator (Maraio, Hall & Taylor 2022) to the real **TR1 South** shear catalogue, at $N_{\rm side} = 64$.
- **All six tomographic bins combined** into one sample (21.9 galaxies/arcmin^2), because single bins (3–4 galaxies/arcmin^2) are much noisier.
- Compared QML with two other estimators on the same data: **NaMaster** (pseudo-C_ℓ) and **heracles**, run directly on the catalogue.
- The patch is small, 0.9% of the sky (about 20° across), so the spectra are measured in bands of $\Delta\ell = 20$.

1. Pixels removed

- Many pixels on the edge of the survey, and along gaps inside it, are only partly covered. They contain few galaxies, so they are much noisier than the rest: up to ∼13,000 times the average.
- QML assumes **every pixel has the same noise**, so these pixels ruined the QML result.
- We keep only pixels that are **at least 50% covered**. This removes 96 of 538 pixels, but only 5% of the galaxies.
- With the cut, QML works. heracles gives almost the same answer with or without the cut, because it already down-weights partly covered pixels.

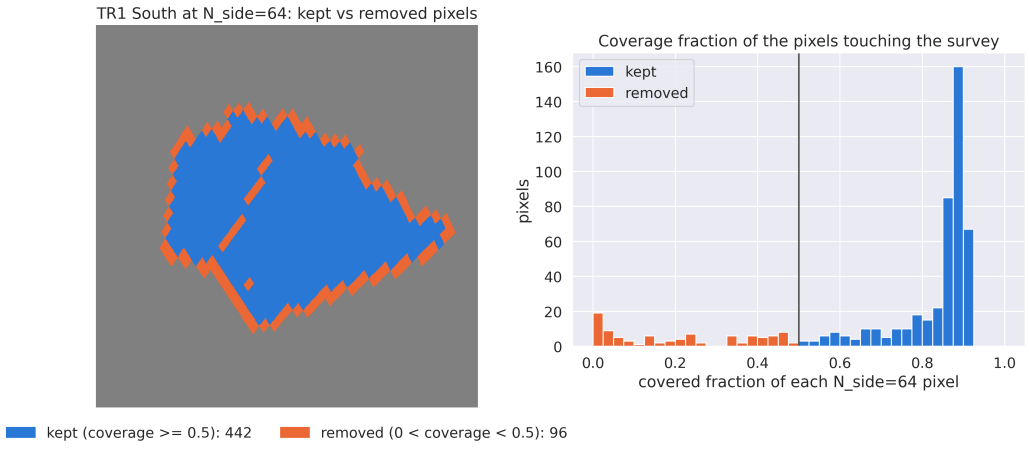


Figure 1: Left: pixels kept (blue) and removed (orange). Right: how much of each pixel is covered; the line is the 50% cut.

2. E-modes: all three estimators agree

QML, NaMaster and heracles all follow the expected (fiducial) spectrum.

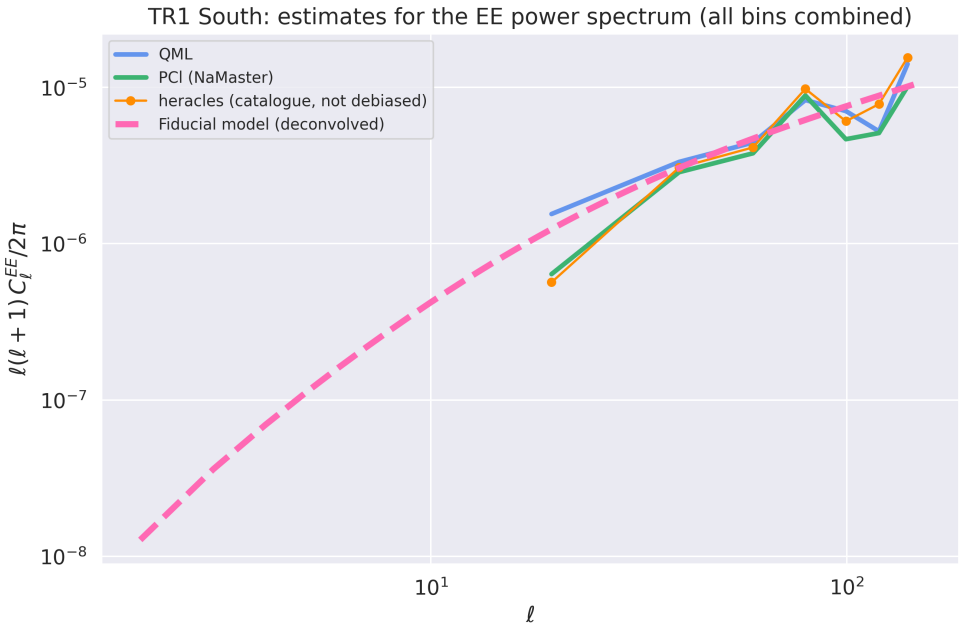


Figure 2: TR1 South E-mode power spectrum from QML, NaMaster and heracles, with the fiducial model.

3. QML has smaller error bars

- The plot shows NaMaster's error bars divided by QML's. Above 1, QML is more precise.
- On large scales ($\ell \approx 20$), QML's errors are **about 3 times smaller for E-modes** and **about 28 times smaller for B-modes**.
- The two become similar by $\ell \approx 100$, as the QML paper predicts: QML's advantage is on large scales.

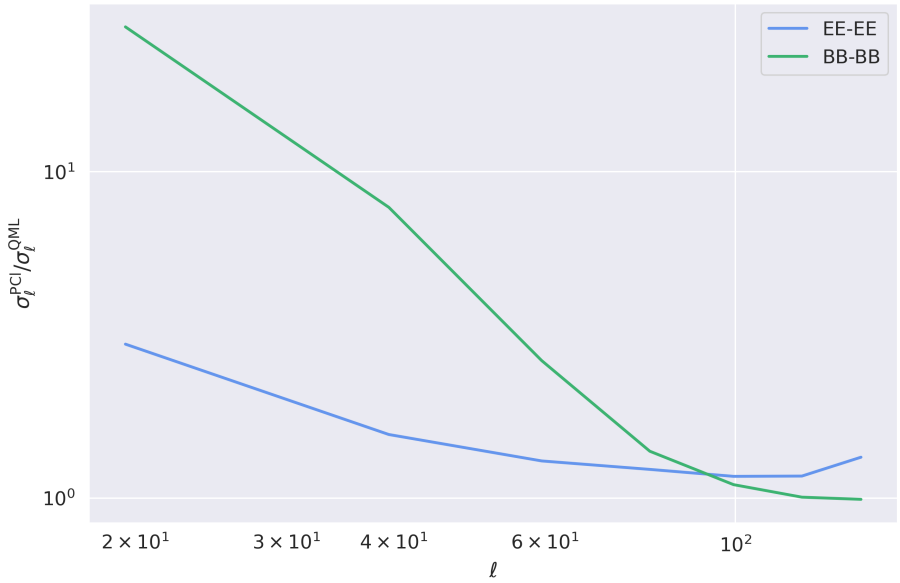


Figure 3: Ratio of NaMaster to QML error bars, for E-modes (blue) and B-modes (green).

4. B-modes

- Weak lensing produces essentially no B-modes, so the measured B-modes should just be noise. Anything above the noise points to leakage or a systematic.
- On large scales ($\ell \lesssim 50$), **QML stays close to the noise level**, while NaMaster and heracles are about 10 times higher. That excess is E-mode power leaking into B through the small, irregular mask, which QML avoids.
- At $\ell \gtrsim 110$ all three estimators are somewhat above the noise. Because all three see it, it is in the data or the pixelisation, not in QML. It is still an open question.

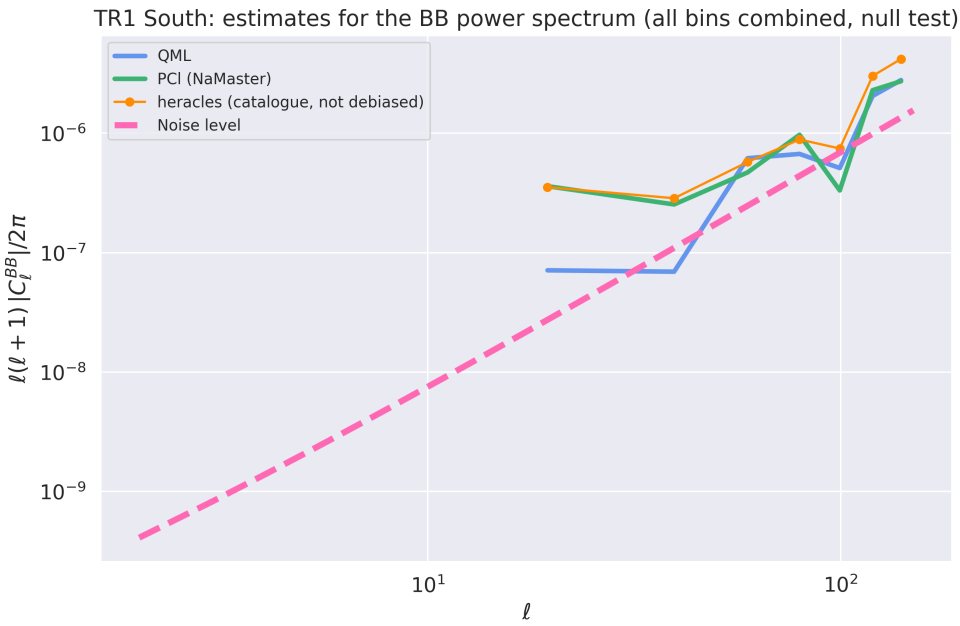


Figure 4: TR1 South B-mode power spectrum from QML, NaMaster and heracles, with the noise level.

Next steps

- **Variable noise in QML:** give QML the noise of each pixel instead of one average value. Then no pixels need removing, and QML handles the coverage the way heracles already does.
- **Per-bin analysis:** repeat for each tomographic bin separately.